



An AI system that stops vendor payment fraud before the money moves — built on Microsoft Azure AI

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Current status: This is a proposed idea backed by research and planning — the full product has not been built yet. A small basic code scaffold has been tested to prove the approach works; the complete working system will be built during the hackathon/project timeline.

1 The Idea, in One Paragraph

VendorGuard is an AI system that reads every incoming vendor invoice the way an experienced accounts-payable officer would — comparing it against that specific vendor's own history (bank account used before, tone of communication, typical amount, domain) — and forces a mandatory phone verification before releasing payment whenever something doesn't match. It targets **Business Email Compromise (BEC)**: a fraud where a scammer hijacks or spoofs a real vendor's email thread and slips in a fake "updated bank details" message that looks exactly like normal correspondence.

IN SIMPLE WORDS — socho tumhara ek regular supplier hai jisse tum har mahine usi bank account mein payment karte ho. Achanak email aata hai "hamara account change ho gaya, is naye account mein bhejo" — dikhta bilkul asli jaisa hai. Na koi hacking, na virus — bas ek smart fake email. Yehi BEC hai.

2 Why This Problem Actually Matters

\$3.05B

reported US losses to BEC in 2025
— 2nd largest fraud category the
FBI tracks

94x

greater than reported ransomware
losses the same year

~\$123K

average loss per incident

45%

of small businesses hit go out of
business within 6 months

Small and mid-sized businesses are hit hardest — too big for the owner to personally review every payment, too small to afford enterprise fraud-prevention software or a dedicated treasury team.

3 Why Nothing Already Solves This

Existing approach	Why it falls short
Enterprise AP fraud tools (Medius, AppZen, Tipalti)	Priced and built for large treasury departments — out of reach for a 10–50 person business.
Email security filters (DMARC, spam filters)	Catch spoofed domains, not a genuinely compromised mailbox sending a legitimate-looking message mid-thread.
Manual "call to verify" policy	The one control that actually works is just a line on a policy document — nobody enforces it consistently under deadline pressure.

The gap: nothing sits between the invoice and the payment step for smaller businesses, reasoning about the specific transaction before the money actually moves.

4 The Solution — 4 AI Agents Working Together

01

Extraction Agent

Pulls vendor, amount, bank details, requester out of the invoice/email.

02

History Agent

Compares new details against this vendor's own fingerprint (bank account, tone, domain, cadence).

03

Reasoning Agent

Weights signals together, writes a plain-English risk rationale — not a black-box score.

04

Verification Agent

On high risk, holds payment and surfaces the vendor's on-file, verified callback number — never the one in the email.

IN SIMPLE WORDS — isse ek 4-step factory line jaisa socho: pehla step data padhta hai, dusra purane record se compare karta hai, teesra sochta hai "risky hai ya nahi," aur chautha payment ko rok ke sahi verification karwata hai.

5 Built Entirely on Microsoft Technology 100% MICROSOFT / AZURE STACK

Microsoft / Azure Service	What It Does
Azure AI Document Intelligence	Microsoft's own prebuilt "invoice" model — automatically reads vendor name, amount, and bank details straight out of a PDF or scanned invoice.
Azure OpenAI Service	Microsoft's cloud-hosted AI reasoning engine — weighs all the detected signals together and writes a human-readable risk explanation.
Azure SQL Database	Stores each vendor's history/"fingerprint," so every new invoice can be compared against it.
Azure Static Web Apps	Hosts the live dashboard where risk scores are displayed, on Microsoft's own cloud.
Microsoft Teams	Sends an instant alert to the finance team the moment a high-risk invoice is flagged.

MENTOR KO YEH ZAROOR BATANA — "Sir, humne jaan-boojhkar har jagah Microsoft ki hi technology use ki hai — data nikalne se lekar AI reasoning tak, aur dashboard hosting tak — taaki yeh idea genuinely Microsoft ke ecosystem ka fayda uthaye, sirf naam ke liye Azure likha hua na ho."

6 Why This Idea Is Unique — What I Thought Differently

Most student/hackathon projects: another AI chatbot, healthcare app, or basic CRUD tool — easy to build but forgettable.

This idea: targets a real, quantifiable, unglamorous-but-massive financial process failure that almost no student project touches.

- **Not a simple rules engine:** a plain rule ("flag any bank change") creates too many false alarms, since vendors sometimes legitimately change accounts. This system weighs multiple weak signals together — tone shift, timing, urgency, amount — the way a skeptical human actually reasons.
- **Explainable by design:** the system never just says "blocked" — it writes out exactly why, so a human trusts and acts on the hold immediately instead of overriding it.
- **The verification number is never taken from the suspicious message** — only from a pre-verified contact record. This one design decision is what makes the control actually work in the real world.

7 Real-World Case Studies — This Is Already Happening

WHY INCLUDE THESE — mentor ko yeh dikhana taaki unhe lage ki yeh koi imaginary problem nahi, balki duniya bhar mein — aur India mein bhi — abhi ho raha hai.

Google & Facebook — \$122 million (2013–2015)

A Lithuanian man, Evaldas Rimasauskas, set up a fake company impersonating Quanta Computer, a real Taiwan-based hardware supplier both tech giants worked with. Using forged invoices and corporate stamps, he convinced finance staff to wire payments to accounts he controlled over roughly two years. Arrested in 2017, sentenced to five years in prison.

Sources: U.S. DOJ (justice.gov), NPR, Sophos

Shivganga Drillers Pvt Ltd, Indore — ₹3.72 crore (2025)

This Indore-based company was making a legitimate international payment to its real US vendor, Innovex International. Fraudsters spoofed the

vendor's email and redirected the payment. Cyber Police later recovered the funds — but the pattern is exactly what VendorGuard is built to catch.

Source: The420.in

Pune engineering/manufacturing firm — ₹73.18 lakh (2026)

A fake vendor identity and a forged quotation for machinery deceived a Pune-based firm — the same core pattern VendorGuard is designed to catch at the invoice-review stage.

Source: The420.in

RKR Gold, Tamil Nadu — ₹16 crore

A Tamil Nadu jewellery exporter had a \$2.57 million gold deal with a Singapore buyer. Fraudsters used an email address almost identical to the real one (a single extra letter) to redirect the payment. Investigated by the CBI.

Source: Hindustan Times (via PressReader)

8 What's Ready, What's Still Pending

Being upfront about this: the full product has not been built yet. The idea, research, and a basic code scaffold are ready to demonstrate that the approach works — the complete working system will be built during the hackathon/project timeline.

- **Ready:** complete research, real case studies, problem-solution fit, and a small test scaffold showing the basic logic of all 4 agents working end-to-end.
- **Pending:** connecting the real Azure AI Document Intelligence and Azure OpenAI services, polishing the dashboard design, and real-world testing.
- **Plan is already clear:** exactly which Azure service goes where has already been decided (see Section 5) — so the build phase should move fast.

9 Honest Limitations

- This is currently a proof-of-concept idea — not a claim of production-grade fraud detection accuracy.
- It doesn't replace human judgment — it enforces the one verification step that already works, automatically.
- Real-world deployment would need testing on real (consented) invoice data from a pilot business.

10 Where This Could Go Next

- **Target customer:** bookkeeping and outsourced-CFO firms serving multiple SME clients at once — one integration reaches many end customers.
- **Future roadmap:** direct inbox integration (Outlook/Gmail), accounting-software plug-ins (QuickBooks/Tally), automated telephony-based callback verification.

Prepared as briefing notes for idea approval / mentor discussion. Statistics: FBI Internet Crime Complaint Center (IC3), 2025 report. Case studies paraphrased from public reporting — sources listed with each case. This is a proposed idea; the full product has not yet been built.